Method 2

Viewing in the direction of x-axis, the required area is the area of the region bounded between $y=\sqrt{3}x$ and $x+y\sqrt{3}=4$, $y=0$ and $y=\sqrt{3}$. So, it can be obtained by applying the formula

$$\int_c^d (x_R - x_L)\,dy$$

Here, $c=0$, $d=\sqrt{3}$, x_R is the x-value on the tangent $x+y\sqrt{3}=4$ and x_L is the x-value on the normal $y=\sqrt{3}x$.

$$\therefore \text{ Area required } = \int_c^d (x_R - x_L)\,dy = \int_0^{\sqrt{3}} \left(\left(4 - y\sqrt{3}\right) - \frac{y}{\sqrt{3}} \right) dy$$

$$= \left[\left(4y - \frac{y^2}{2}\sqrt{3} \right) - \frac{y^2}{2\sqrt{3}} \right]_0^{\sqrt{3}}$$

$$= 4\sqrt{3} - \frac{3}{2}\sqrt{3} - \frac{3}{2\sqrt{3}} = 2\sqrt{3}.$$

Working rule for finding area of the region bounded by $y=f_1(x)$, $y=f_2(x)$, the lines $x=a$ and $x=b$, where $a<b$:

Draw an arbitrary line parallel to y-axis cutting the plane region. First, find the y-coordinate of the point where the line enters the region. Call it y_{ENTRY}. Next, find the y-coordinate of the point where the line exits the region. Call it y_{EXIT}. Both y_{ENTRY} and y_{EXIT} can be found from the equations of the bounding curves. Then, the required area is given by $\int_a^b [y_{EXIT} - y_{ENTRY}]\,dx$.

Working rule for finding area of the region bounded by $x=g_1(y)$, $x=g_2(y)$, the lines $y=c$ and $y=d$, where $c<d$:

Draw an arbitrary line parallel to x-axis cutting the plane region.

First, find the x-coordinate of the point where the line enters the region. Call it x_{ENTRY} .

Next, find the x-coordinate of the point where the line exits the region. Call it x_{EXIT}. Both x_{ENTRY} and x_{EXIT} can be found from the equations of the bounding curves. Then, the required area is given by $\int_c^d [x_{EXIT} - x_{ENTRY}]\,dy$.

EXERCISE 9.8

1. Find the area of the region bounded by $3x-2y+6=0$, $x=-3$, $x=1$ and x-axis.
2. Find the area of the region bounded by $2x-y+1=0$, $y=-1$, $y=3$ and y-axis.
3. Find the area of the region bounded by the curve $2+x-x^2+y=0$, x-axis, $x=-3$ and $x=3$.
4. Find the area of the region bounded by the line $y=2x+5$ and the parabola $y=x^2-2x$.
5. Find the area of the region bounded between the curves $y=\sin x$ and $y=\cos x$ and the lines $x=0$ and $x=\pi$.
6. Find the area of the region bounded by $y=\tan x$, $y=\cot x$ and the lines $x=0$, $x=\frac{\pi}{2}$, $y=0$.
7. Find the area of the region bounded by the parabola $y^2=x$ and the line $y=x-2$.

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